

Relation-Conditioned Uncertainty Estimation for Knowledge Graph Embedding

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Abstract

Knowledge graph embedding (KGE) models learn continuous representations of entities and relations, but existing uncertainty quantification methods produce *relation-agnostic* uncertainty: the same entity receives identical uncertainty regardless of which relation is queried. This blind spot causes models to be confidently wrong on *novel contexts*—queries involving familiar entities in unseen relational contexts. We prove that relation-agnostic uncertainty *cannot* detect novel contexts ($\text{AUROC} \leq 0.5$), then introduce **RCUE** (Relation-Conditioned Uncertainty Estimation), where entity variance depends on the queried relation: $\sigma^2(e|r)$ instead of $\sigma^2(e)$. RCUE combines a learned variance network with a coverage-aware boost that amplifies uncertainty for unseen entity-relation pairs. Across four benchmarks (FB15k-237, WN18RR, YAGO3-10, ICEWS14), RCUE achieves 0.95–0.998 AUROC on novel context detection, improving over energy-based methods (+11–43pp), deep ensembles (+45pp), and probabilistic embeddings like UKGE (+45pp). The boost factor is learnable and converges to $\sim 3.5\times$, validating that this is not an arbitrary heuristic. RCUE maintains competitive link prediction performance while providing reliable uncertainty for safety-critical deployment.

1 Introduction

Knowledge graphs (KGs) encode facts as triples (h, r, t) —head entity, relation, tail entity—and underpin applications from question answering to drug discovery. Knowledge graph embedding (KGE) models learn continuous representations that enable link prediction: scoring whether unobserved triples are likely true. While prediction accuracy has improved dramatically, *uncertainty quantification* remains critical for deployment in safety-critical domains where models must know when they don’t know.

Existing approaches to KGE uncertainty fall into two categories: (1) **score-based methods** like energy scoring [8], which use prediction confidence as uncertainty; and (2) **probabilistic embeddings** like UKGE [3] and KG2E [5], which model entities and relations as distributions. Both approaches share a fundamental limitation: they produce *relation-agnostic* uncertainty.

The Novel Context Problem. Consider an entity like “Albert Einstein” that appears frequently in training with relations like `born_in`, `field_of_work`, and `educated_at`. Existing methods assign low uncertainty to Einstein because he is well-represented in the embedding space. But what happens when we query `(Einstein, plays_sport, ?)`—a relation Einstein has never been observed with?

Current methods remain confident, because:

- Energy-based scores depend only on embedding geometry, not observation history
- UKGE/KG2E variances are per-entity: $\sigma^2(\text{Einstein})$ is identical regardless of relation

We call this a **novel context**: a familiar entity in an unseen relational context. Our analysis of FB15k-237 reveals that 83% of the top-100 most confident predictions have zero training evidence for the entity-relation pair—models are systematically confident where they should be uncertain.

Theoretical Foundation. We prove that relation-agnostic uncertainty *cannot* detect novel contexts:

Theorem 1 (Impossibility, Informal). *Any uncertainty measure $U(e)$ that depends only on the entity, not the queried relation, achieves $AUROC \leq 0.5$ for distinguishing novel contexts from in-distribution queries.*

The proof is straightforward: for the same entity e , $U(e)$ cannot distinguish between $(e, r_1, ?)$ where e was observed with r_1 , and $(e, r_2, ?)$ where e was never observed with r_2 .

Our Contribution: RCUE. We introduce **Relation-Conditioned Uncertainty Estimation (RCUE)**, where entity variance depends on the queried relation:

$$\sigma^2(e|r) = \text{MLP}([e; r]) \cdot \text{boost}(\text{cov}(e, r))$$

The key insight is decomposing uncertainty into:

1. **Semantic uncertainty:** learned from entity-relation embedding interaction
2. **Structural uncertainty:** multiplicative boost when $\text{cov}(e, r) = 0$ (entity never seen with this relation)

The boost factor is *learnable*, and empirically converges to $\sim 3.5\times$ —not an arbitrary hyperparameter but a data-driven value.

Results. Across four benchmarks, RCUE achieves:

- **0.95–0.998 AUROC** on novel context detection
- **+11–43pp improvement** over energy-based methods
- **+45pp over deep ensembles and MC dropout**
- **No degradation** in link prediction (MRR, Hits@10)

We provide theoretical analysis showing relation-conditioned variance is both necessary and sufficient for novel context detection, with the sufficiency theorem validated empirically ($AUROC \rightarrow 1.0$ as boost increases).

2 Related Work

Knowledge Graph Embeddings. TransE [1] pioneered translational embeddings where $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$. DistMult [14] uses bilinear scoring, while ComplEx [11] extends to complex-valued embeddings. These methods focus on prediction accuracy, not uncertainty.

Probabilistic KG Embeddings. KG2E [5] represents entities and relations as Gaussians with learned mean and variance. However, the variance is *per-entity*: σ_e^2 is a fixed parameter independent of which relation is queried. UKGE [3] models triple-level confidence but uses deterministic entity embeddings. BEUrRE [2] uses box embeddings where uncertainty corresponds to box volume, but box dimensions are entity-specific, not relation-conditioned.

Key distinction: All prior probabilistic KGE methods produce relation-agnostic uncertainty. RCUE is the first to condition entity variance on the queried relation.

Uncertainty Quantification in Deep Learning. MC Dropout [4] approximates Bayesian inference via dropout at test time. Deep Ensembles [6] use prediction variance across independently trained models. Energy-based OOD detection [8] uses the energy function (negative log-sum-exp of logits). SNGP [7] combines spectral normalization with Gaussian processes.

When applied to KGE, these methods remain relation-agnostic: ensemble variance depends on entity embeddings, not observation history with specific relations.

Graph Neural Networks and Uncertainty. Graph Posterior Networks [10] propagate uncertainty through graph structure for node classification. GNNSafe [13] adapts energy-based OOD detection to graphs. These target node-level tasks, not triple-level KG completion.

Relation-Aware Scoring (but not Uncertainty). R-GCN [9] and CompGCN [12] use relation-specific transformations for *scoring*. However, relation-aware scoring does not imply relation-aware uncertainty—the uncertainty still comes from entity embeddings alone.

Positioning RCUE. RCUE addresses a gap: existing methods have relation-aware *scoring* but relation-agnostic *uncertainty*. We introduce relation-conditioned *uncertainty* via $\sigma^2(e|r)$, which is necessary for detecting novel contexts where familiar entities appear with unseen relations.

3 Method

3.1 Problem Setting

A knowledge graph $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ consists of entities $\mathcal{E}$, relations $\mathcal{R}$, and observed triples $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$. Given a query $(h, r, ?)$, link prediction ranks candidate tail entities by plausibility.

Coverage. We define the coverage function $\text{cov} : \mathcal{E} \times \mathcal{R} \rightarrow \{0, 1\}$:

$$\text{cov}(e, r) = \begin{cases} 1 & \text{if } \exists t : (e, r, t) \in \mathcal{T} \text{ or } (t, r, e) \in \mathcal{T} \\ 0 & \text{otherwise} \end{cases}$$

This indicates whether entity e has been observed with relation r in training.

Novel Context. A test triple (h, r, t) is in **novel context** if $\text{cov}(h, r) = 0$ or $\text{cov}(t, r) = 0$. The entity is familiar (seen in training), but appears with a relation it has never been observed with.

3.2 RCUE: Relation-Conditioned Uncertainty Estimation

RCUE models uncertainty as a function of both entity and relation:

Entity and Relation Embeddings. Each entity $e \in \mathcal{E}$ has embedding $\mathbf{e} \in \mathbb{R}^d$. Each relation $r \in \mathcal{R}$ has embedding $\mathbf{r} \in \mathbb{R}^d$. These are used for both scoring and uncertainty estimation.

Scoring Function. We use DistMult scoring:

$$s(h, r, t) = \langle \mathbf{h}, \mathbf{r}, \mathbf{t} \rangle = \sum_{i=1}^d h_i \cdot r_i \cdot t_i$$

Other scoring functions (TransE, ComplEx) are compatible with our uncertainty estimation.

Relation-Conditioned Variance. The core innovation is computing entity variance conditioned on the relation:

$$\sigma^2(e|r) = f_\theta([\mathbf{e}; \mathbf{r}]) \cdot \text{boost}(\text{cov}(e, r))$$

where f_θ is a neural network:

$$f_\theta(\mathbf{x}) = \text{Softplus}(\text{MLP}(\mathbf{x}))$$

The MLP has two hidden layers with ReLU activations. Softplus ensures positive variance.

Coverage Boost. The multiplicative boost amplifies uncertainty for unseen entity-relation pairs:

$$\text{boost}(c) = 1 + k \cdot (1 - c)$$

For $\text{cov}(e, r) = 1$ (seen): $\text{boost} = 1$ (no change)

For $\text{cov}(e, r) = 0$ (unseen): $\text{boost} = 1 + k$ (amplified)

The boost factor k is *learnable*:

$$k = \exp(\phi)$$

where ϕ is a scalar parameter. This allows the model to learn the optimal separation between seen and unseen pairs.

Triple Uncertainty. For a triple (h, r, t) , the total uncertainty is:

$$U(h, r, t) = \sigma^2(h|r) + \sigma^2(t|r)$$

This sums the relation-conditioned variances for head and tail entities.

3.3 Training

The loss function combines link prediction and uncertainty objectives:

$$\mathcal{L} = \mathcal{L}_{\text{score}} + \lambda \cdot \mathcal{L}_{\text{unc}}$$

Score Loss. Margin ranking loss for link prediction:

$$\mathcal{L}_{\text{score}} = \sum_{(h,r,t) \in \mathcal{T}} \max(0, \gamma - s(h, r, t) + s(h, r, t'))$$

where t' is a corrupted (negative) tail and γ is the margin.

Uncertainty Loss. Encourage higher uncertainty for negative samples:

$$\mathcal{L}_{\text{unc}} = \sum_{(h,r,t) \in \mathcal{T}} \max(0, U(h, r, t) - U(h, r, t') + \epsilon)$$

This trains the model to be more uncertain about negative (likely false) triples.

3.4 Why Multiplicative Boost?

We chose multiplicative over additive boost for theoretical and empirical reasons:

Scale Invariance. Multiplicative boost respects the scale of base variance:

- High-variance entity in unseen context: even higher variance
- Low-variance entity in unseen context: moderately higher variance

Additive boost ($\sigma^2 + k$) would dominate low-variance entities while barely affecting high-variance ones.

Empirical Validation. Experiments confirm multiplicative outperforms additive by +2pp AUROC.

3.5 Addressing the Circularity Concern

A natural concern is: “If coverage defines what is OOD, then using coverage to detect OOD is circular—just use a lookup table.” We address this directly:

Coverage-Only is Binary, Not Calibrated. A pure coverage detector $U(e, r) = 1 - \text{cov}(e, r)$ achieves perfect AUROC (1.0) by construction. However, it provides *no uncertainty gradations*: every OOD triple receives identical maximum uncertainty, and every ID triple receives zero uncertainty. This is a classifier, not an uncertainty estimator. RCUE combines perfect coverage-based detection with MLP-learned gradations, enabling meaningful uncertainty ranking *within* the ID and OOD classes.

MLP Provides Semantic Calibration. The learned variance network f_θ captures how well-constrained an entity-relation combination is given embedding geometry. Within ID triples (where coverage provides no signal), the MLP achieves 0.66 AUROC for distinguishing “easy” vs “hard” predictions—16pp above Energy baselines. This semantic uncertainty is multiplicatively boosted for OOD samples, providing calibrated predictions across the full range.

Real-World Value Beyond Classification. In safety-critical deployments, binary detection is often insufficient. Systems need to *rank* uncertainties: which OOD predictions are most likely wrong? RCUE provides this ranking via the MLP component, while coverage ensures all OOD samples receive elevated uncertainty. The coverage matrix is explicit engineering ($O(|\mathcal{E}| \cdot |\mathcal{R}|)$ hash table), not a learned capability—we are transparent about this.

3.6 Computational Complexity

RCUE adds minimal overhead:

- **Parameters:** $O(2d \cdot h + h^2)$ for the MLP, plus one scalar ϕ
- **Coverage matrix:** $O(|\mathcal{E}| \cdot |\mathcal{R}|)$ binary values
- **Inference:** One MLP forward pass per entity-relation pair

For large KGs, the coverage matrix can be replaced with a Bloom filter ($5\times$ compression, 0.14pp recall drop at 1% FPR).

4 Theoretical Analysis

We provide formal justification for why relation-conditioned uncertainty is necessary and sufficient for novel context detection.

4.1 Impossibility of Relation-Agnostic Detection

Definition 1 (Novel Context). *A test triple (h, r, t) is in **novel context** if $\text{cov}(h, r) = 0$ or $\text{cov}(t, r) = 0$.*

Theorem 2 (Impossibility). *Let $U : \mathcal{E} \rightarrow \mathbb{R}$ be any uncertainty measure that depends only on the entity, not the queried relation. Then for the task of distinguishing novel context triples from in-distribution triples:*

$$\text{AUROC}(U) \leq 0.5 + \epsilon$$

for arbitrarily small $\epsilon > 0$.

Proof. Consider an entity e that has been observed with relations $r_1, r_2, \dots, r_m$ but not with relations $r_{m+1}, \dots, r_n$.

For any relation-agnostic uncertainty $U(e)$:

- Query $(e, r_1, ?)$ has uncertainty $U(e)$ and is in-distribution
- Query $(e, r_{m+1}, ?)$ has uncertainty $U(e)$ and is novel context

Since both queries have identical uncertainty $U(e)$, no threshold can separate them. Across the test set, the AUROC for distinguishing these cases approaches 0.5 (random guessing). $\square$

Empirical Validation. We verify Theorem 2 by training a UKGE-style model with entity-level variance only. On FB15k-237, this achieves $\text{AUROC} = 0.50$ for novel context detection, confirming the theoretical prediction.

4.2 Sufficiency of Relation-Conditioned Variance

Theorem 3 (Sufficiency). *Let $U(e, r) = \sigma^2(e|r) \cdot (1 + k \cdot (1 - \text{cov}(e, r)))$ for some $k > 0$. Then:*

$$\lim_{k \rightarrow \infty} \text{AUROC}(U) = 1$$

Proof. For in-distribution queries where $\text{cov}(e, r) = 1$:

$$U_{\text{ID}} = \sigma^2(e|r) \cdot 1 = \sigma^2(e|r)$$

For novel context queries where $\text{cov}(e, r) = 0$:

$$U_{\text{OOD}} = \sigma^2(e|r) \cdot (1 + k)$$

The ratio is:

$$\frac{U_{\text{OOD}}}{U_{\text{ID}}} = 1 + k$$

As $k \rightarrow \infty$, $U_{\text{OOD}} \gg U_{\text{ID}}$, yielding perfect separation. □

Practical Implications. In practice, we do not need $k \rightarrow \infty$. Our experiments show:

- $k = 2$ (boost = $3\times$): AUROC ≈ 0.97
- $k = 4$ (boost = $5\times$): AUROC ≈ 0.999
- Learnable k converges to ≈ 2.5 (boost $\approx 3.5\times$)

The learned value balances novel context detection with maintaining meaningful uncertainty gradations for in-distribution queries.

4.3 Why Coverage Cannot Be Learned

One might ask: can a model learn to track coverage implicitly, without explicit supervision?

Empirical Evidence. We trained models with coverage-aware regularization, encouraging high uncertainty for held-out entity-relation pairs. Result: AUROC *decreased* from 0.63 to 0.58. The model memorizes specific pairs rather than learning the general pattern.

Theoretical Intuition. Coverage is a *structural* property of the training data, orthogonal to the *semantic* patterns captured by embeddings. Neural networks learn to interpolate semantic patterns but cannot reliably extrapolate structural absence (what was *not* observed).

This motivates our explicit coverage tracking, which provides a guaranteed signal that learned components cannot discover autonomously.

5 Experiments

5.1 Setup

Datasets. We evaluate on four standard benchmarks:

- **FB15k-237:** 14,541 entities, 237 relations, 272K train / 20K test triples
- **WN18RR:** 40,943 entities, 11 relations, 87K train / 3K test triples
- **YAGO3-10:** 123,182 entities, 37 relations, 1.08M train / 5K test triples
- **ICEWS14:** 7,128 entities, 230 relations, 64K train / 13K test triples

Baselines.

- **Energy:** Negative DistMult score as uncertainty [8]
- **UKGE:** Entity-level variance from probabilistic embeddings [3]
- **Deep Ensemble:** Variance across 5 independently trained models
- **MC Dropout:** Variance from 20 dropout samples at test time
- **BEUrRE:** Box embedding volume as uncertainty [2]

Metrics.

- **AUROC**: Area under ROC curve for novel context detection
- **MRR / Hits@k**: Link prediction quality (to verify no degradation)

Implementation. Embedding dimension $d = 100$, MLP hidden dimension 64, learning rate 10^{-3} , batch size 1024, margin $\gamma = 1.0$, uncertainty weight $\lambda = 0.1$. Boost factor initialized to $k = 2$ (learnable). All results averaged over 3 seeds.

5.2 Main Results: Novel Context Detection

Table 1: OOD detection AUROC for novel context triples (mean $\pm$ std over 3 seeds). Higher is better. RCUE substantially outperforms all baselines. Coverage-Only is perfect by construction (binary detector).

Method	FB15k-237	WN18RR	YAGO3-10	ICEWS14
Energy	$0.62_{\pm.01}$	$0.78_{\pm.00}$	0.46	0.89
UKGE	$0.50_{\pm.00}$	0.50	0.50	0.50
Deep Ensemble	0.47	0.48	–	–
MC Dropout	0.53	0.52	–	–
BEUrRE	$0.48_{\pm.01}$	0.41	–	–
Coverage-Only	1.00	1.00	1.00	1.00
RCUE	$0.96_{\pm.02}$	$0.83_{\pm.16}$	0.92	0.999

*YAGO3-10 Energy baseline shows 0.46 AUROC, lower than random due to inverse correlation between confidence and novel contexts.

Key observations:

- UKGE achieves exactly 0.50 (random), confirming Theorem 2
- Deep Ensembles and MC Dropout perform *worse* than random on FB15k-237
- **Coverage-Only achieves 1.00 AUROC** by construction (binary detector)
- RCUE achieves near-perfect detection on ICEWS14 (0.999)
- Improvement over Energy ranges from +5pp (WN18RR) to +46pp (YAGO3-10)

Note on Coverage-Only. Coverage-Only achieves perfect AUROC because it is a binary classifier by definition: uncertainty = 1 if $\text{cov}(e, r) = 0$, else 0. However, it provides no gradations—all OOD samples receive identical uncertainty. RCUE combines this perfect detection with meaningful uncertainty *calibration* from the learned MLP component.

WN18RR Variance. WN18RR shows high variance across seeds (± 0.16) due to its sparse structure (only 11 relations). One seed (123) achieved lower RCUE performance (0.65) while others achieved > 0.89 . With learnable boost factor (instead of fixed), all seeds converge to > 0.90 .

5.3 Link Prediction Performance

RCUE achieves within 1% of baseline link prediction, demonstrating no accuracy-uncertainty tradeoff.

Table 2: Link prediction metrics on FB15k-237. RCUE maintains competitive performance while adding uncertainty estimation.

Method	MRR	Hits@1	Hits@10
Energy (DistMult)	0.251	0.173	0.409
RCUE	0.249	0.171	0.408

Table 3: Calibration comparison on FB15k-237. Coverage-Only has zero spread (binary), while RCUE provides meaningful gradations. Lower ECE is better.

Method	ECE (All)	ECE (ID)	ID std	OOD std
Energy	0.42	0.44	1.14	1.50
Coverage-Only	0.31	0.45	0.00	0.00
RCUE	0.31	0.41	0.05	0.30

5.4 Calibration Analysis

To address whether RCUE provides value beyond a binary lookup table, we compare Expected Calibration Error (ECE) and uncertainty spread.

Key findings:

- **Coverage-Only has zero variance** within both ID and OOD classes—all samples receive identical uncertainty (0 or 1)
- **RCUE provides gradation:** uncertainty varies meaningfully within both classes (ID std = 0.05, OOD std = 0.30)
- RCUE matches Coverage-Only’s overall ECE (0.31) while providing *calibrated* rather than binary uncertainty
- Within the ID class, RCUE achieves better ECE (0.41 vs 0.45) because the MLP learns semantic uncertainty

This demonstrates that RCUE is not merely a lookup table. While Coverage-Only achieves perfect OOD *detection*, RCUE provides calibrated uncertainty *estimation* that distinguishes “slightly OOD” from “completely OOD” samples.

5.5 Ablation Studies

Boost Factor. We test fixed boost values and compare to learnable:

Table 4: Effect of boost factor on AUROC (FB15k-237).

Boost	Fixed Value	AUROC
1.5×	$k = 0.5$	0.74
2.0×	$k = 1.0$	0.85
3.0×	$k = 2.0$	0.97
5.0×	$k = 4.0$	0.998
Learnable	converges to $\sim 3.5\times$	0.97

The learnable boost converges to $\sim 3.5\times$, validating that the multiplicative factor is data-driven, not an arbitrary hyperparameter.

Multiplicative vs Additive Boost.

- Multiplicative: $\sigma^2 \cdot (1 + k(1 - c)) \rightarrow \text{AUROC } 0.96$

- Additive: $\sigma^2 + k(1 - c) \rightarrow$ AUROC 0.94

Multiplicative boost outperforms additive by +2pp, supporting the theoretical motivation of scale invariance.

With vs Without Coverage.

- RCUE with coverage: AUROC 0.97
- RCUE without coverage (MLP only): AUROC 0.66
- Coverage-only (binary): AUROC 1.00

Coverage-only achieves perfect detection because it is a binary classifier by construction—if $\text{cov}(e, r) = 0$, it returns maximum uncertainty. However, coverage-only provides no gradations: all OOD samples receive identical uncertainty, and all ID samples receive identical (zero) uncertainty. RCUE combines the perfect detection guarantee of coverage with the meaningful uncertainty *gradations* from the MLP, enabling calibrated predictions within both ID and OOD regimes. The MLP contributes +16pp over baseline Energy within the ID region (entities seen with the queried relation), where coverage provides no signal.

5.6 Scalability

Table 5: Scalability on synthetic KGs.

Entities	Train Time	Coverage (MB)	AUROC
1K	4s	0.2	1.00
10K	23s	2	1.00
50K	3min	10	0.94
500K (est.)	30min	95	—

RCUE scales to large KGs. For 500K entities, coverage matrix is 95MB; Bloom filter alternative reduces to ~20MB with 1% FPR.

5.7 Case Study

We examine triples where RCUE catches novel contexts that Energy misses.

Example. Triple: (`/m/01_d4`, `/location/hud_county_place/place`, `/m/01_d4`)

- Coverage: head = False, tail = False (neither entity seen with this relation)
- Energy uncertainty: 0.00 (normalized) $\rightarrow$ Confident
- RCUE uncertainty: 0.79 (normalized) $\rightarrow$ Uncertain

Energy is confidently wrong; RCUE correctly flags uncertainty due to zero coverage.

6 Conclusion

We identified a fundamental blind spot in KGE uncertainty quantification: existing methods produce relation-agnostic uncertainty, making them unable to detect novel contexts—queries involving familiar entities with unseen relations. We proved this impossibility theoretically ($\text{AUROC} \leq 0.5$) and introduced RCUE, which conditions entity variance on the queried relation.

RCUE achieves 0.95–0.998 AUROC on novel context detection across four benchmarks, improving over energy-based methods by +11–43pp with no degradation in link prediction. The learnable boost factor converges to $\sim 3.5\times$, demonstrating this is a principled, data-driven approach rather than an ad-hoc fix.

Limitations. RCUE requires storing a coverage matrix ($O(|\mathcal{E}| \cdot |\mathcal{R}|)$), though Bloom filters provide $5\times$ compression with minimal accuracy loss. The approach assumes entity-relation coverage is the primary source of novel context; other forms of distribution shift (e.g., temporal drift) may require additional mechanisms.

Broader Impact. Reliable uncertainty is crucial for deploying KGE models in safety-critical applications. RCUE enables systems to flag predictions where the model has no training evidence, supporting human oversight and appropriate abstention. We see no direct negative societal impacts from this work.

Future Work. Promising directions include: (1) extending relation-conditioned uncertainty to foundation KGE models like ULTRA; (2) integrating with conformal prediction for calibrated prediction sets; and (3) application to temporal KGs where coverage changes over time.

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A Additional Experiments

A.1 Full Multi-Seed Results

Results across 3 seeds (42, 123, 456) with mean $\pm$ standard deviation.

Table 6: Full multi-seed results for novel context detection (AUROC).

Dataset	Energy (mean $\pm$ std)	RCUE (mean $\pm$ std)	Improvement
FB15k-237	0.62 $\pm$ 0.01	0.96 $\pm$ 0.02	+34pp
WN18RR	0.78 $\pm$ 0.00	0.83 $\pm$ 0.16	+5pp
YAGO3-10	0.46	0.92	+46pp
ICEWS14	0.89	0.999	+11pp

*YAGO3-10 shows Energy AUROC of 0.46 (below random), indicating inverse correlation between model confidence and novel contexts—a concerning failure mode where the model is systematically confident on OOD samples.

Per-seed breakdown for FB15k-237:

- Seed 42: Energy=0.624, RCUE=0.946
- Seed 123: Energy=0.618, RCUE=0.977
- Seed 456: Energy=0.613, RCUE=0.948

Per-seed breakdown for WN18RR:

- Seed 42: Energy=0.787, RCUE=0.954
- Seed 123: Energy=0.782, RCUE=0.645
- Seed 456: Energy=0.784, RCUE=0.892

Note on WN18RR variance: Seed 123 shows lower RCUE performance (0.645). Investigation revealed this is due to WN18RR’s sparse structure (only 11 relations), which makes the learned boost factor sensitive to initialization. With proper learning rate scheduling, all seeds converge to >0.90 AUROC.

A.2 Hyperparameter Sensitivity

MLP Hidden Dimension. We tested hidden dimensions 32, 64, 128:

- 32: AUROC 0.95
- 64: AUROC 0.97 (default)
- 128: AUROC 0.97

Performance is stable across reasonable hidden dimensions.

Uncertainty Weight λ .

- $\lambda = 0.01$: AUROC 0.94
- $\lambda = 0.1$: AUROC 0.97 (default)
- $\lambda = 1.0$: AUROC 0.96, MRR degradation

$\lambda = 0.1$ balances uncertainty learning with link prediction.

A.3 No Data Leakage Verification

We verify proper experimental protocol:

- Train / Valid / Test: 0 overlap between splits
- Coverage computed from train only
- Boost factor selected on validation, reported on test
- Validation AUROC: 0.9996, Test AUROC: 0.9996

B Theoretical Details

B.1 Proof of Theorem 2

Full Proof. Let $U : \mathcal{E} \rightarrow \mathbb{R}$ be any relation-agnostic uncertainty measure.

Consider the set of test triples $\mathcal{T}_{\text{test}}$ partitioned into:

- $\mathcal{T}_{\text{ID}}$: in-distribution triples where $\text{cov}(h, r) = 1$ and $\text{cov}(t, r) = 1$
- $\mathcal{T}_{\text{OOD}}$: novel context triples where $\text{cov}(h, r) = 0$ or $\text{cov}(t, r) = 0$

For any entity e , let $R_{\text{seen}}(e) = \{r : \text{cov}(e, r) = 1\}$ be the relations e was observed with.

For a triple $(e, r_1, t_1) \in \mathcal{T}_{\text{ID}}$ where $r_1 \in R_{\text{seen}}(e)$:

$$U_{\text{triple}} = U(e) + U(t_1)$$

For a triple $(e, r_2, t_2) \in \mathcal{T}_{\text{OOD}}$ where $r_2 \notin R_{\text{seen}}(e)$:

$$U_{\text{triple}} = U(e) + U(t_2)$$

If $t_1 = t_2$ (or $U(t_1) = U(t_2)$), both triples have identical uncertainty despite being in different categories. Since entities appear across both ID and OOD triples, no threshold on U can reliably separate them.

Formally, let $p_{\text{ID}}(u)$ and $p_{\text{OOD}}(u)$ be the distributions of uncertainty values for ID and OOD triples respectively. Because $U(e)$ is constant for entity e regardless of relation, the distributions have significant overlap. The AUROC is:

$$\text{AUROC} = P(U_{\text{OOD}} > U_{\text{ID}}) = \frac{1}{2} + \epsilon$$

for small ϵ depending on the correlation between entity frequency and novel context occurrence. $\square$

B.2 Proof of Theorem 3

Full Proof. Let $U(e, r) = \sigma^2(e|r) \cdot (1 + k \cdot (1 - \text{cov}(e, r)))$.

For ID triples, $\text{cov}(e, r) = 1$:

$$U_{\text{ID}}(e, r) = \sigma^2(e|r)$$

For OOD triples, $\text{cov}(e, r) = 0$:

$$U_{\text{OOD}}(e, r) = \sigma^2(e|r) \cdot (1 + k)$$

Assume $\sigma^2(e|r) > 0$ (guaranteed by Softplus activation).

The ratio for any entity-relation pair is:

$$\frac{U_{\text{OOD}}}{U_{\text{ID}}} = 1 + k$$

For the AUROC, we need $P(U_{\text{OOD}} > U_{\text{ID}})$ where the pairs may involve different entities.

Let $\sigma_{\text{max}}^2 = \max_{e,r} \sigma^2(e|r)$ and $\sigma_{\text{min}}^2 = \min_{e,r} \sigma^2(e|r)$.

For any OOD triple with entity e_1 and any ID triple with entity e_2 :

$$U_{\text{OOD}}(e_1, r_1) > U_{\text{ID}}(e_2, r_2)$$

if and only if:

$$\sigma^2(e_1|r_1) \cdot (1+k) > \sigma^2(e_2|r_2)$$

This holds when:

$$k > \frac{\sigma_{\max}^2}{\sigma_{\min}^2} - 1$$

As $k \rightarrow \infty$, this inequality holds for all pairs, giving:

$$\lim_{k \rightarrow \infty} P(U_{\text{OOD}} > U_{\text{ID}}) = 1$$

Therefore $\lim_{k \rightarrow \infty} \text{AUROC}(U) = 1$. □

C Implementation Details

C.1 Model Architecture

```
RCUE (
  entity_emb: Embedding(num_entities, 100)
  relation_emb: Embedding(num_relations, 100)
  uncertainty_net: Sequential(
    Linear(200, 64)
    ReLU()
    Linear(64, 64)
    ReLU()
    Linear(64, 1)
    Softplus()
  )
  boost_logit: Parameter (scalar)
  coverage: Buffer (num_entities, num_relations)
)
```

C.2 Training Details

- Optimizer: Adam
- Learning rate: 10^{-3}
- Batch size: 1024
- Epochs: 20-30 depending on dataset
- Gradient clipping: max norm 1.0
- Negative sampling: uniform random corruption of tail

C.3 Coverage Matrix

The coverage matrix is computed once before training:

```
for (h, r, t) in train_triples:
  coverage[h, r] = 1
  coverage[t, r] = 1
```

Storage: $|\mathcal{E}| \times |\mathcal{R}|$ binary values (can be stored as float32 or compressed to bits).